

Hoang Anh (Ben) NGUYEN

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RESEARCH SUMMARY

I am a PhD student in Geophysics at Colorado School of Mines. My thesis develops quantum computing algorithms for solving PDEs and inverse problems. I also work on large-scale seismic multicomponent elastic full-waveform inversion with GPU-accelerated solvers, including distributed acoustic sensing data, and on physics-informed neural networks.

EDUCATION

Ph.D. in Geophysics Major: Geophysics, Minor: Computer Science Colorado School of Mines (CSM) Thesis: Solving inverse problems with quantum computing	Aug 2023 – May 2027 Golden, CO, USA
Postgraduate Diploma in Earth System Physics Major: Solid Earth International Centre for Theoretical Physics (ICTP) - UNESCO Thesis: Ambient noise tomography beneath the Banda Arc	Sept 2022 – Aug 2023 Trieste, Italy
Master of Engineering Physics Major: Computational Physics Hanoi University of Science and Technology (HUST) Thesis: Structural simulation of MgSiO ₃ under compression	May 2021 – Sept 2023 Hanoi, Vietnam
Erasmus Exchange Major: Physics of Complex Systems Polytechnic University of Turin (POLITO)	Feb 2022 – Jul 2022 Torino, Italy
Bachelor of Engineering Physics Major: Computational Physics Talent Honours Program – HUST Thesis: Computational modeling of microstructure of magnesium silicate	Sept 2016 – April 2021 Hanoi, Vietnam

CURRENT WORK & PUBLICATIONS

- [1] Nguyen, H.A., Tura, A. Joint multi-parameter elastic full-waveform inversion using distributed acoustic sensing and multi-component seismic data. Submitted to *Computers & Geosciences* (2026). arxiv.org/abs/2607.01649
- [2] Nguyen, H.A., Vashisth, D., Tura, A. Hybrid quantum-classical finite-basis physics-informed neural network for wave propagation and full waveform inversion. Under review at *IEEE Transactions on Geoscience & Remote Sensing* (2026). arxiv.org/abs/2606.01110
- [3] Nguyen, H.A., Manu-Marfo, D., & Aoudia, A. Ambient noise tomography beneath the Banda basin reveals new insights into the Arc-Continent collision zone. Under review at *Geophys. Res. Lett.* (2026). [10.22541/es-soar.176790173.31128656/v1](https://doi.org/10.22541/es-soar.176790173.31128656/v1)
- [4] Plan, E.L.C.V.M., Phan, H., Nguyen, H.A. *et al.* Numerical simulation on structural and topological transitions of GeO₂ liquid under compression. *Eur. Phys. J. B* **99**, 20 (2026). [10.1140/epjb/s10051-026-01143-0](https://doi.org/10.1140/epjb/s10051-026-01143-0)
- [5] Nguyen, H.A., Tura, A. Seismic traveltime inversion with quantum annealing. *Scientific Reports* **15**, 17984 (2025). [10.1038/s41598-025-01188-8](https://doi.org/10.1038/s41598-025-01188-8)
- [6] Lai, D.V., Nguyen, S.H., Nguyen, H.A. *et al.* Tailoring hydrogen storage performance of Mg–Mg₂Ni alloys: synergistic effects of composition and phase formation with first-principles insights. *RSC Advances* **15**, 31240–31254 (2025). [10.1039/D5RA04356E](https://doi.org/10.1039/D5RA04356E)
- [7] Nguyen, V.H., Nguyen, H.A. Crystallisation of liquid silica under compression: a molecular dynamics simulation. *Pramana - J Phys* **98**, 142 (2024). [10.1007/s12043-024-02839-7](https://doi.org/10.1007/s12043-024-02839-7)
- [8] Nguyen, H.A., Nguyen, V.H. Study of the structure of MgSiO₃ system under compression by using ring statistics and Voronoi analysis. *Phys. Scr.* **98**, 045919 (2023). [10.1088/1402-4896/acc5b7](https://doi.org/10.1088/1402-4896/acc5b7)
- [9] Nguyen, V.H., Nguyen, H.A., Iitaka, T., Mai, T.L. Computer simulation of phosphate-silicate and calcium phosphate-silicate systems. *Phys. Scr.* **98**, 065704 (2023). [10.1088/1402-4896/acd4fb](https://doi.org/10.1088/1402-4896/acd4fb)
- [10] Nguyen, H.A., Nguyen, S., Nguyen, V.H. Pressure-induced glassy networks of enstatite (MgSiO₃) and forsterite (Mg₂SiO₄). *VNU J. Sci. Math. - Phys.* **39**, 1 (2023). [10.25073/2588-1124/vnumap.4767](https://doi.org/10.25073/2588-1124/vnumap.4767)

- [11] Pham, T.H.H., Doan, H.H., Ta, Q.M., Mai, T.L., Nguyen, H.A. Some results of seismic travel-time reflection tomography study. *Petrovietnam Journal* **10**, 4–16 (2021). [10.47800/PVJ.2021.10-01](https://doi.org/10.47800/PVJ.2021.10-01)
- [12] Nguyen, H.S., Nguyen, H.A., Pham, H.K., Iitaka, T., Nguyen, V.H. Topology of SiO_x units and glassy network of magnesium silicate glass under densification: correlation between radial distribution function and bond angle distribution. *Modelling Simul. Mater. Sci. Eng.* **28**, 065007 (2020). [10.1088/1361-651X/ab9bb4](https://doi.org/10.1088/1361-651X/ab9bb4)
- [13] Nguyen, H.S., Nguyen, H.A. Structural simulation of Mg_2SiO_4 under compression. *VNU J. Sci. Math. - Phys.* **36**, 4 (2020). [10.25073/2588-1124/vnumap.4484](https://doi.org/10.25073/2588-1124/vnumap.4484)

RESEARCH EXPERIENCE

Research Assistant

Department of Geophysics, CSM

Aug 2023 – Present

Advisor: Prof. Tura, A.

- Design and implement classical and hybrid quantum–classical PINNs for seismic inverse problems, integrating finite-basis parameterizations with machine learning based optimization.
- Investigate quantum algorithms for optimization and wave equation solvers in seismic modeling.
- Build a GPU-accelerated multicomponent elastic FWI framework, including the distributed acoustic sensing (DAS) data, with perfectly matched layer absorbing boundary conditions, achieving up to $100\times$ speedup relative to baseline multi-core CPU implementations.

Research Assistant

Earth System Physics, ICTP

May 2023 – Aug 2023

Advisors: Prof. Aoudia, A. &

Dr. Manu-Marfo, D.

- Developed MPI-parallelized ambient noise tomography workflows using four years of continuous seismic data to extract empirical Green's functions and perform nonlinear inversion of surface-wave dispersion for shear-wave velocity model reconstruction down to 200 km depth, including resolution analysis.

Research Assistant

Department of Computational Physics, HUST

Aug 2018 – July 2023

Advisor: Prof. Nguyen, V.H.

- Developed parallel molecular dynamics codes in C on RIKEN's [Fugaku](#) supercomputer to simulate silicate, germanate, and phosphate glasses and liquids under compression, characterizing pressure-induced coordination and topological transitions.
- Performed first-principles density functional theory calculations using [Quantum ESPRESSO](#), evaluating electronic structure and formation enthalpies with plane-wave pseudopotential methods.

WORK EXPERIENCE

Geophysics R&D Research Intern

[Shell](#): global integrated energy company

May 2026 – Aug 2026

Houston, TX

- Handled undulating interfaces in wave propagation and FWI using finite-difference and spectral-element methods.

Geophysics R&D Research Intern

[TGS](#): energy data and analytics company

May 2025 – Aug 2025

Houston, TX

- Developed GPU-accelerated seismic wave solvers for high-performance computing environments.
- Modeled Q-attenuation and compensation in isotropic and anisotropic (VTI/TTI) media using Devito Pro.
- Formulated an anisotropic attenuation model on fully staggered grids.
- Implemented Q-elastic gradient computation for reverse time migration (RTM) and FWI.

Geophysics R&D Research Intern

[VPI](#): national energy research institute

Sept 2021 – Feb 2022

Hanoi, Vietnam

- Implemented 3D seismic ray-tracing algorithms for travel-time modeling in heterogeneous velocity models.
- Applied non-linear travel-time tomography techniques for subsurface velocity reconstruction.

INTERNATIONAL TALKS AND PRESENTATIONS

- [1] IMAGE26 (oral): Nguyen, H.A., Tura, A. Full waveform inversion with hybrid quantum finite basis physics-informed neural network Aug 2026
Houston, TX
- [2] EAGE26 (workshop): Tura, A., Yermakov, R., Nguyen, H.A., Sabbagh, R. Leveraging multicomponent OBN acquisitions for improved PP-PS imaging and inversion Jun 2026
Aberdeen, Scotland
- [3] [IASPE-IAGA25](#) (poster): Manu-Marfo, D., Nguyen, H.A., Aoudia, A. Ambient noise tomography beneath the Banda basin reveals new insights into the arc-continent collision zone Sep 2025
Lisbon, Portugal
- [4] IMAGE25 (workshop): Nguyen, H.A. Optimization with quantum annealing method Aug 2025
Houston, TX
- [5] IMAGE25 (oral): Nguyen, H.A., Tura, A. Crosswell travelttime inversion using a quantum computing method Aug 2025
Houston, TX

SERVICE AND MENTORING

- [1] Peer-review service: Geophysical Journal International (Oxford), [IMAGE](#) (SEG), Acta Geophysica
- [2] PhD application mentoring: mentored 3 students attending PhD programs in the US

HONORS AND AWARDS

- [1] Fully funded scholarship for postgraduate program at ICTP 2023
- [2] VEF 2.0 Program recommended candidate 2023: The [VEF 2.0](#) Program is conducted by the Fellows and Scholars of the Vietnam Education Foundation (VEF) – an independent U.S. Federal Government agency created by the U.S. Congress
- [3] Erasmus scholarship for master exchange students at POLITO 2022
- [4] Fully funded scholarship for the master program at HUST 2021, 2022
- [5] Certificate of Merit from School of Engineering Physics for undergraduates: Excellent Student in Fall Semester 2017, Spring Semester 2018, Fall Semester 2018, Spring Semester 2019
- [6] The 20th Vietnam National Student Physics Olympiad 2018: Second Prize
- [7] Lawrence S. Ting Scholarship 2017 for undergraduates
- [8] Vietnam Robot National Contest 2019: team member of HUST

OTHER CERTIFICATIONS

- [1] Deep Learning Specialization from DeepLearning.AI ([Certificate](#))
- [2] Quantum Computing and Quantum Information from IBM ([Certificate](#))

TECHNICAL SKILLS

Programming: Python, Julia, C, C++, Fortran, MATLAB

Machine Learning Frameworks: JAX, PyTorch, TensorFlow, scikit-learn

Geophysical Modeling: Anisotropic elastic FWI, RTM, DAS FWI, ambient noise tomography, quantum PDE solvers

Optimization: Adjoint-state methods, quantum annealing

HPC: Slurm, MPI, OpenMP/OpenACC, GPU computing